

Equations with Fractions

Name
Date
Course

Solve

$$\textcircled{1} \quad 2x + \frac{1}{6} = -\frac{1}{2} + \frac{15x}{18}$$

$$\frac{36x}{18} + \frac{3}{18} = -\frac{9}{18} + \frac{15x}{18}$$

$$\frac{36x+3}{18} = \frac{-9+15x}{18}$$

$$36x+3 = -9+15x$$

$$21x = -12$$

$$\boxed{x = -\frac{4}{7}}$$

$$\textcircled{3} \quad -3x + \frac{7}{15} = \frac{2}{3} + \frac{x}{5}$$

$$\frac{-45x}{15} + \frac{7}{15} = \frac{10}{15} + \frac{3x}{15}$$

$$\frac{-45x+7}{15} = \frac{10+3x}{15}$$

$$-45x+7 = 10+3x$$

$$-48x = 3$$

$$x = -\frac{3}{48}$$

$$\boxed{x = -\frac{1}{16}}$$

$$\textcircled{2} \quad \frac{1}{4} + \frac{3z}{8} = 4z - \frac{1}{2}$$

$$\frac{2}{8} + \frac{3z}{8} = \frac{32z}{8} - \frac{4}{8}$$

$$\frac{2+3z}{8} = \frac{32z-4}{8}$$

$$2+3z = 32z-4$$

$$6 = 29z$$

$$\boxed{\frac{6}{29} = z}$$

$$\textcircled{4} \quad \frac{3}{4} - \frac{z}{2} = 3z - 2$$

$$\frac{3}{4} - \frac{2z}{4} = \frac{12z}{4} - \frac{8}{4}$$

$$\frac{3-2z}{4} = \frac{12z-8}{4}$$

$$3-2z = 12z-8$$

$$11 = 14z$$

$$\boxed{\frac{11}{14} = z}$$

$$\textcircled{5} \quad x+4 = \frac{3x}{5} - \frac{x}{2}$$

$$\frac{10x}{10} + \frac{40}{10} = \frac{6x}{10} - \frac{5x}{10}$$

$$\frac{10x+40}{10} = \frac{x}{10}$$

$$10x+40 = x$$

$$40 = -9x$$

$$\boxed{-\frac{40}{9} = x}$$

$$\textcircled{6} \quad \frac{3y}{7} - \frac{3y}{14} = 2y + \frac{1}{28}$$

$$\frac{8y}{28} - \frac{6y}{28} = \frac{56y}{28} + \frac{1}{28}$$

$$\frac{2y}{28} = \frac{56y+1}{28}$$

$$2y = 56y+1$$

$$-54y = 1$$

$$\boxed{y = -\frac{1}{54}}$$

$$\textcircled{7} \quad \frac{2}{5y} + \frac{1}{10} = 3 - \frac{4}{y}$$

$$\frac{4}{10y} + \frac{y}{10y} = \frac{30y}{10y} - \frac{40}{10y}$$

$$\frac{4+y}{10y} = \frac{30y-40}{10y}$$

$$4+y = 30y-40$$

$$44 = 29y$$

$$\boxed{\frac{44}{29} = y}$$

$$\textcircled{8} \quad \frac{2}{5x} + \frac{1}{25} = 1 - \frac{2}{x}$$

$$\frac{10}{25x} + \frac{x}{25x} = \frac{25x}{25x} - \frac{50}{25x}$$

$$\frac{10+x}{25x} = \frac{25x-50}{25x}$$

$$10+x = 25x-50$$

$$60 = 24x$$

$$\frac{60}{24} = x$$

$$\boxed{\frac{5}{2} = x}$$

$$\textcircled{9} \quad \frac{3}{7x^2} - \frac{2}{x} = \frac{1}{5x}$$

$$\frac{15}{35x^2} - \frac{70x}{35x^2} = \frac{7x}{35x^2}$$

$$\frac{15-70x}{35x^2} = \frac{7x}{35x^2}$$

$$15-70x=7x$$

$$15=77x$$

$$\boxed{\frac{15}{77} = x}$$

$$\textcircled{10} \quad \frac{1}{7y} - \frac{2}{y^2} = \frac{5}{14y}$$

$$\frac{2y}{14y^2} - \frac{28}{14y^2} = \frac{5y}{14y^2}$$

$$\frac{2y-28}{14y^2} = \frac{5y}{14y^2}$$

$$2y-28=5y$$

$$-28=3y$$

$$\boxed{-\frac{28}{3} = y}$$

$$\star \textcircled{11} \quad z-4 = \frac{45}{z}$$

$$\frac{z^2}{z} - \frac{4z}{z} = \frac{45}{z}$$

$$\frac{z^2-4z}{z} = \frac{45}{z}$$

$$z^2-4z=45$$

$$z^2-4z-45=0$$

$$(z-9)(z+5)=0$$

$$\boxed{z=9} \quad \boxed{z=-5}$$

$$\textcircled{12} \quad z = 10 - \frac{21}{z}$$

$$\frac{z^2}{z} = \frac{10z}{z} - \frac{21}{z}$$

$$\frac{z^2}{z} = \frac{10z-21}{z}$$

$$z^2 = 10z-21$$

$$z^2-10z+21=0$$

$$(z-3)(z-7)=0$$

$$\boxed{z=3} \quad \boxed{z=7}$$

$$\textcircled{13} \quad -\frac{15}{2x} = \frac{x}{1} + \frac{11}{2}$$

$$-\frac{15}{2x} = \frac{2x^2}{2x} + \frac{11x}{2x}$$

$$-\frac{15}{2x} = \frac{2x^2+11x}{2x}$$

$$-15 = 2x^2+11x$$

$$0 = 2x^2+11x+15$$

$$0 = (2x+5)(x+3)$$

$$\boxed{x = -\frac{5}{2}} \quad \boxed{x = -3}$$

$$\textcircled{15} \quad \frac{3x^2}{4} - \frac{1}{1} = \frac{x}{1}$$

$$\frac{3x^2}{4} - \frac{4}{4} = \frac{4x}{4}$$

$$\frac{3x^2-4}{4} = \frac{4x}{4}$$

$$3x^2-4=4x$$

$$3x^2-4x-4=0$$

$$(3x+2)(x-2)=0$$

$$\boxed{x = -\frac{2}{3}} \quad \boxed{x = 2}$$

$$\textcircled{14} \quad \frac{5y}{2} + \frac{10}{y} = -14.5$$

$$\frac{5y^2}{2y} + \frac{20}{2y} = \frac{-29y}{2y}$$

$$\frac{5y^2+20}{2y} = \frac{-29y}{2y}$$

$$5y^2+20=-29y$$

$$5y^2+29y+20=0$$

$$(5y+4)(y+5)=0$$

$$\boxed{y = -\frac{4}{5}} \quad \boxed{y = -5}$$

$$\textcircled{16} \quad \frac{x}{2} + \frac{12}{x} = \frac{5}{1}$$

$$\frac{x^2}{2x} + \frac{24}{2x} = \frac{10x}{2x}$$

$$\frac{x^2+24}{2x} = \frac{10x}{2x}$$

$$x^2+24=10x$$

$$x^2-10x+24=0$$

$$(x-4)(x-6)=0$$

$$\boxed{x=4} \quad \boxed{x=6}$$

$$\textcircled{17} \quad \frac{10}{y^4} = \frac{13}{y^3} - \frac{3}{y^2}$$

$$\frac{10}{y^4} = \frac{13y}{y^4} - \frac{3y^2}{y^4}$$

$$\frac{10}{y^4} = \frac{13y - 3y^2}{y^4}$$

$$10 = 13y - 3y^2$$

$$3y^2 - 13y + 10 = 0$$

$$(3y-10)(y-1) = 0$$

$$\boxed{y = \frac{10}{3}} \quad \boxed{y = 1}$$

$$\textcircled{19} \quad \frac{2}{x-1} + \frac{x-6}{2x^2+5x-7} = \frac{3}{2x+7}$$

$$\frac{2}{x-1} + \frac{x-6}{(2x+7)(x-1)} = \frac{3}{2x+7}$$

$$\frac{(2x+7)2}{(2x+7)(x-1)} + \frac{x-6}{(2x+7)(x-1)} = \frac{3}{(2x+7)(x-1)}$$

$$\frac{4x+14}{(2x+7)(x-1)} + \frac{x-6}{(2x+7)(x-1)} = \frac{3x-3}{(2x+7)(x-1)}$$

$$\frac{5x+8}{(2x+7)(x-1)} = \frac{3x-3}{(2x+7)(x-1)}$$

$$5x+8 = 3x-3$$

$$2x = -11$$

$$\boxed{x = -11/2}$$

$$\textcircled{18} \quad \frac{1}{z^2} + \frac{28}{z^4} = -\frac{11}{z^3}$$

$$\frac{z^2}{z^4} + \frac{28}{z^4} = \frac{-11z}{z^4}$$

$$\frac{z^2+28}{z^4} = \frac{-11z}{z^4}$$

$$z^2+28 = -11z$$

$$z^2+11z+28=0$$

$$(z+7)(z+4)=0$$

$$\boxed{z = -7} \quad \boxed{z = -4}$$

$$\textcircled{20} \quad \frac{3}{x+2} = \frac{x-5}{2x^2+5x+2} + \frac{1}{2x+1}$$

$$\frac{3}{(x+2)} = \frac{x-5}{(2x+1)(x+2)} + \frac{1}{(2x+1)}$$

$$\frac{(2x+1)3}{(2x+1)(x+2)} = \frac{x-5}{(2x+1)(x+2)} + \frac{1}{(2x+1)(x+2)}$$

$$\frac{6x+3}{(2x+1)(x+2)} = \frac{x-5}{(2x+1)(x+2)} + \frac{x+2}{(2x+1)(x+2)}$$

$$\frac{6x+3}{(2x+1)(x+2)} = \frac{2x-3}{(2x+1)(x+2)}$$

$$6x+3 = 2x-3$$

$$4x = -6$$

$$\boxed{x = -\frac{3}{2}}$$

$$\textcircled{21} \quad \frac{6}{y-4} - \frac{2}{y+4} = \frac{-y}{y^2-16}$$

$$\frac{6}{y-4} - \frac{2}{y+4} = \frac{-y}{(y+4)(y-4)}$$

$$\frac{(y+4)6}{(y+4)(y-4)} - \frac{2(y-4)}{(y+4)(y-4)} = \frac{-y}{(y+4)(y-4)}$$

$$\frac{6y+24}{(y-4)(y+4)} - \frac{2y-8}{(y-4)(y+4)} = \frac{-y}{(y+4)(y-4)}$$

$$\frac{4y+32}{(y-4)(y+4)} = \frac{-y}{(y+4)(y-4)}$$

$$4y+32 = -y$$

$$32 = -5y$$

$$\boxed{-\frac{32}{5} = y}$$

$$\textcircled{22} \quad \frac{3y}{y^2-9} = \frac{8}{y-3} + \frac{2}{y+3}$$

$$\frac{3y}{(y-3)(y+3)} = \frac{8}{y-3} + \frac{2}{y+3}$$

$$\frac{3y}{(y-3)(y+3)} = \frac{8(y+3)}{(y-3)(y+3)} + \frac{2(y-3)}{(y-3)(y+3)}$$

$$\frac{3y}{(y-3)(y+3)} = \frac{8y+24}{(y-3)(y+3)} + \frac{2y-6}{(y-3)(y+3)}$$

$$\frac{3y}{(y-3)(y+3)} = \frac{10y+18}{(y-3)(y+3)}$$

$$3y = 10y+18$$

$$-7y = 18$$

$$\boxed{y = -\frac{18}{7}}$$

$$\textcircled{23} \quad \frac{7}{2z^2-10z} + \frac{-1}{z} = \frac{1}{z-5}$$

$$\frac{7}{2z(z-5)} + \frac{-1}{z} = \frac{1}{z-5}$$

$$\frac{7}{2z(z-5)} + \frac{2(z-5)(-1)}{2(z-5)z} = \frac{1}{(z-5)z}$$

$$\frac{7}{2z(z-5)} + \frac{-2z+10}{2z(z-5)} = \frac{2z}{2z(z-5)}$$

$$\frac{-2z+17}{2z(z-5)} = \frac{2z}{2z(z-5)}$$

$$-2z+17 = 2z$$

$$17 = 4z$$

$$\boxed{\frac{17}{4} = z}$$

$$\textcircled{24} \quad \frac{3}{z+3} + \frac{-2}{2z^2+6z} = \frac{1}{2z}$$

$$\frac{3}{z+3} + \frac{-2}{2z(z+3)} = \frac{1}{2z}$$

$$\frac{(2z)3}{(2z)(z+3)} + \frac{-2}{2z(z+3)} = \frac{1}{2z(z+3)}$$

$$\frac{6z}{2z(z+3)} + \frac{-2}{2z(z+3)} = \frac{z+3}{2z(z+3)}$$

$$\frac{6z-2}{2z(z+3)} = \frac{z+3}{2z(z+3)}$$

$$6z-2 = z+3$$

$$5z = 5$$

$$\boxed{z = 1}$$

$$\begin{aligned} (25) \quad \frac{1}{x-1} + \frac{x}{x-8} &= \frac{-7}{x^2-9x+8} \\ \frac{1}{x-1} + \frac{x}{x-8} &= \frac{-7}{(x-8)(x-1)} \\ \frac{(x-8) \cdot 1}{(x-8)(x-1)} + \frac{(x-1)x}{(x-1)(x-8)} &= \frac{-7}{(x-8)(x-1)} \\ \frac{x-8}{(x-8)(x-1)} + \frac{x^2-x}{(x-1)(x-8)} &= \frac{-7}{(x-8)(x-1)} \\ \frac{x^2-8}{(x-1)(x-8)} &= \frac{-7}{(x-8)(x-1)} \\ x^2-8 &= -7 \\ x^2 &= 1 \\ \sqrt{x^2} &= \pm\sqrt{1} \\ x &= \pm 1 \end{aligned}$$

$x \neq 1$ or $x = -1$

$$\begin{aligned} (27) \quad 1 + \frac{2}{z-1} &= \frac{6}{z^2-z} \\ \frac{1(z(z-1))}{1(z(z-1))} + \frac{2(z)}{(z-1)(z)} &= \frac{6}{z(z-1)} \\ \frac{z(z-1)}{z(z-1)} + \frac{2z}{z(z-1)} &= \frac{6}{z(z-1)} \\ \frac{z^2-2}{z(z-1)} + \frac{2z}{z(z-1)} &= \frac{6}{z(z-1)} \\ \frac{z^2+z-2}{z(z-1)} &= \frac{6}{z(z-1)} \\ z^2+z-2 &= 6 \\ z^2+z-8 &= 0 \\ (z+3)(z-2) &= 0 \\ z &= -3 \quad z = 2 \end{aligned}$$

$$\begin{aligned} (26) \quad \frac{-4}{x^2+14x+45} &= \frac{1}{x+9} + \frac{x}{x+5} \\ \frac{-4}{(x+9)(x+5)} &= \frac{1}{x+9} + \frac{x}{x+5} \\ \frac{-4}{(x+9)(x+5)} &= \frac{(x+5) \cdot 1}{(x+5)(x+9)} + \frac{(x+9)x}{(x+9)(x+5)} \\ \frac{-4}{(x+9)(x+5)} &= \frac{x+5}{(x+9)(x+5)} + \frac{x^2+9x}{(x+9)(x+5)} \\ \frac{-4}{(x+9)(x+5)} &= \frac{x^2+10x+5}{(x+9)(x+5)} \\ -4 &= x^2+10x+5 \\ 0 &= x^2+10x+9 \\ 0 &= (x+9)(x+1) \\ x &\neq -9 \quad x = -1 \end{aligned}$$

$$\begin{aligned} (28) \quad -\frac{20}{z^2+z} + 1 &= \frac{2}{z+1} \\ \frac{-20}{z(z+1)} + 1 &= \frac{2}{z+1} \\ \frac{-20}{z(z+1)} + \frac{1 \cdot z(z+1)}{z(z+1)} &= \frac{2 \cdot z}{(z+1)z} \\ \frac{-20}{z(z+1)} + \frac{z(z+1)}{z(z+1)} &= \frac{2z}{z(z+1)} \\ \frac{-20}{z(z+1)} + \frac{z^2+z}{z(z+1)} &= \frac{2z}{z(z+1)} \\ \frac{z^2+z-20}{z(z+1)} &= \frac{2z}{z(z+1)} \\ z^2+z-20 &= 2z \\ z^2-z-20 &= 0 \\ (z-5)(z+4) &= 0 \\ z &= 5 \quad z = -4 \end{aligned}$$

$$\begin{aligned} (29) \quad \frac{x+3}{16x^2-9} + \frac{2}{4x+3} &= \frac{1}{4x-3} \\ \frac{x+3}{(4x-3)(4x+3)} + \frac{2}{4x+3} &= \frac{1}{4x-3} \\ \frac{x+3}{(4x-3)(4x+3)} + \frac{(4x-3) \cdot 2}{(4x-3)(4x+3)} &= \frac{1 \cdot (4x+3)}{(4x-3)(4x+3)} \\ \frac{x+3}{(4x-3)(4x+3)} + \frac{8x-6}{(4x-3)(4x+3)} &= \frac{4x+3}{(4x-3)(4x+3)} \\ \frac{9x-3}{(4x-3)(4x+3)} &= \frac{4x+3}{(4x-3)(4x+3)} \\ 9x-3 &= 4x+3 \\ 5x &= 6 \\ x &= \frac{6}{5} \end{aligned}$$

$$\begin{aligned} (31) \quad \frac{9}{x^2-5x+4} &= \frac{1}{x-1} + \frac{3}{x-4} \\ \frac{9}{(x-4)(x-1)} &= \frac{1}{x-1} + \frac{3}{x-4} \\ \frac{9}{(x-4)(x-1)} &= \frac{1(x-4)}{(x-1)(x-4)} + \frac{3(x-1)}{(x-4)(x-1)} \\ \frac{9}{(x-4)(x-1)} &= \frac{x-4}{(x-1)(x-4)} + \frac{3x-3}{(x-4)(x-1)} \\ \frac{9}{(x-4)(x-1)} &= \frac{4x-7}{(x-1)(x-4)} \\ 9 &= 4x-7 \\ 16 &= 4x \\ 4 &\neq x \quad \text{No Solution} \end{aligned}$$

$$\begin{aligned} (30) \quad -\frac{2x+7}{9x^2-49} + \frac{5}{3x+7} &= \frac{1}{3x-7} \\ -\frac{2x+7}{(3x-7)(3x+7)} + \frac{5}{3x+7} &= \frac{1}{3x-7} \\ -\frac{2x+7}{(3x-7)(3x+7)} + \frac{5(3x-7)}{(3x+7)(3x-7)} &= \frac{1(3x+7)}{(3x-7)(3x+7)} \\ -\frac{2x+7}{(3x-7)(3x+7)} + \frac{15x-35}{(3x+7)(3x-7)} &= \frac{3x+7}{(3x-7)(3x+7)} \\ \frac{13x-42}{(3x-7)(3x+7)} &= \frac{3x+7}{(3x-7)(3x+7)} \\ 13x-42 &= 3x+7 \\ 10x &= 49 \\ x &= \frac{49}{10} = 4.9 \end{aligned}$$

$$\begin{aligned} (32) \quad \frac{x}{x+4} - \frac{1}{x+8} &= \frac{4x}{x^2+12x+32} \\ \frac{x}{x+4} - \frac{1}{x+8} &= \frac{4x}{(x+4)(x+8)} \\ \frac{x(x+8)}{(x+4)(x+8)} - \frac{1(x+4)}{(x+8)(x+4)} &= \frac{4x}{(x+4)(x+8)} \\ \frac{x^2+8x}{(x+4)(x+8)} - \frac{x+4}{(x+8)(x+4)} &= \frac{4x}{(x+4)(x+8)} \\ \frac{x^2+7x-4}{(x+4)(x+8)} &= \frac{4x}{(x+4)(x+8)} \\ x^2+7x-4 &= 4x \\ x^2+3x-4 &= 0 \\ (x+4)(x-1) &= 0 \\ x &= -4 \quad x = 1 \end{aligned}$$

$$* \textcircled{33} \frac{3}{z+2} = \frac{2}{z-1} + \frac{1}{z+4} \quad \textcircled{34} \frac{6}{z-5} - \frac{2}{z-1} = \frac{4}{z+3}$$

$$\textcircled{35} \frac{3}{y+1} - \frac{5}{y-1} = \frac{-2}{y+3} \quad \textcircled{36} -\frac{8}{y+2} + \frac{5}{y+7} = \frac{-3}{y+1}$$

$$* \textcircled{37} \frac{-20}{x^2-16x-36} + \frac{x+2}{x-18} = \frac{1}{x+2}$$

$$\frac{-20}{(x+2)(x-18)} + \frac{x+2}{x-18} = \frac{1}{x+2}$$

$$\frac{-20}{(x+2)(x-18)} + \frac{(x+2)(x+2)}{(x+2)(x-18)} = \frac{1(x-18)}{(x+2)(x-18)}$$

$$\frac{-20}{(x+2)(x-18)} + \frac{x^2+4x+4}{(x+2)(x-18)} = \frac{x-18}{(x+2)(x-18)}$$

$$\frac{x^2+4x-16}{(x+2)(x-18)} = \frac{x-18}{(x+2)(x-18)}$$

$$x^2+4x-16 = x-18$$

$$x^2+3x+2 = 0$$

$$(x+1)(x+2) = 0$$

$$\boxed{x=-1} \quad x \neq -2$$

$$\textcircled{38} \frac{z+5}{z-1} = \frac{15}{z+2} + \frac{13}{z^2+z-2}$$

$$\frac{z+5}{z-1} = \frac{15}{z+2} + \frac{13}{(z+2)(z-1)}$$

$$\frac{(z+5)(z+2)}{(z-1)(z+2)} = \frac{15(z+2)}{(z+2)(z-1)} + \frac{13}{(z+2)(z-1)}$$

$$\frac{z^2+7z+10}{(z-1)(z+2)} = \frac{15z+30}{(z+2)(z-1)} + \frac{13}{(z+2)(z-1)}$$

$$\frac{z^2+7z+10}{(z-1)(z+2)} = \frac{15z-2}{(z+2)(z-1)}$$

$$z^2+7z+10 = 15z-2$$

$$z^2-8z+12 = 0$$

$$(z-6)(z-2) = 0$$

$$\boxed{z \neq 6} \quad \boxed{z=2}$$

$$\textcircled{39} \frac{3}{x-2} - \frac{x-1}{x-4} = \frac{-5}{x^2+2x-8} \quad \textcircled{40} \frac{z+3}{z-10} - \frac{2}{z+1} = \frac{3z^2+7z+20}{z^2-9z-10}$$